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General Problem: On Quality of Life

Specific Problem: To compare the quality of life of working mothers in the context of remote and onsite working conditions by administering the WHOQOL-BREF (abbreviated version of WHOQOL)

Basic Concept

Quality of life is defined as “an individual's perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards, and concerns” (World Health Organization, 1997). “Quality of life (QOL) is the extent to which objective human needs are fulfilled in relation to personal or group perceptions of subjective well-being” (Costanza et al., 2007).

“Quality of life is the degree to which a person enjoys the important possibilities of his or her life” (Quality of Life Research Unit, University of Toronto, n.d.). Quality of life (QoL) refers to an individual's overall well-being and satisfaction with life, taking into account physical health, psychological state, social relationships, and environment. It is a multidimensional and subjective concept influenced by cultural, social, and personal factors. According to the World Health Organization (1997), QoL is “an individual's perception of their position in life in the context of the culture and value systems in which they live.” Costanza et al. (2007) emphasize the balance between objective needs and subjective well-being, while the University of Toronto’s model focuses on enjoying life’s important possibilities, such as personal growth and social belonging.

Importance of Quality of Life

Quality of life (QOL) is an essential indicator in health and medical research, as it provides valuable insight into how illness, treatment, and broader social circumstances affect an

individual's well-being and daily functioning. Understanding QOL is critical for improving symptom relief, optimising care, and enhancing rehabilitation for patients across different age groups and cultural backgrounds. Assessing QOL not only helps identify a range of problems that may affect patients, some of which may persist long after treatment, but also supports better medical decision-making by highlighting the prognostic value of QOL in predicting treatment success and survival.

In clinical practice and research, QOL assessment is increasingly used to evaluate the outcomes of interventions, guide the allocation of healthcare resources, and inform patients about the potential benefits and risks of various treatment options. By focusing on patient-reported experiences, QOL measures help shift the emphasis from purely biomedical outcomes to the broader impact of health on personal fulfilment, social relationships, and daily activities. This patient-centred approach enables healthcare providers and policymakers to better understand the consequences of health conditions and interventions, ultimately fostering more effective, person-centred care and informed health policy.

Description and Development of WHOQOL

The WHOQOL (World Health Organization Quality of Life) is a standardized questionnaire developed by the World Health Organization to assess individuals' perceptions of their quality of life. The WHOQOL-100 is the original, comprehensive version of the World Health Organization's Quality of Life assessment tool. It consists of 100 items that measure an individual's subjective perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards and concerns. It is a 100-question assessment that currently exists in directly comparable forms in 29 language versions.

The WHOQOL-100 have many uses, including use in medical practice, research, health policy making and in assessing the effectiveness and relative merits of different treatments. They can also be used to assess variation in quality of life across different cultures, to compare subgroups within the same culture and to measure change across time in response to change in life circumstances.

The WHOQOL was developed in the early 1990s by the World Health Organization in collaboration with 15 international field centres. Each contributes to the identification of key facets of life relevant to quality of life assessment. These teams helped define and operationalize these facets and proposed items for inclusion. An initial pilot version with 236 items was created and tested on over 300 participants across different health conditions. Based on the data collected from this multicentric pilot study, 100 statistically reliable, valid and conceptually sound items were selected to form the refined WHOQOL-100 field trial version, ensuring cross-cultural validity and comprehensiveness. It was refined to ensure strong internal consistency, test-retest reliability, content validity, and cross-cultural comparability. Ultimately, it produced a profile across 6 domains and 24 facets.

Physical Capacity. This domain includes pain and discomfort, energy and fatigue, as well as sleep and rest.

Psychological. This domain covers positive feelings, thinking, learning, memory and concentration, along with self-esteem, body image and appearance, and negative feelings.

Level of Independence. This domain involves mobility, activities of daily living, dependence on medication or treatments, and work capacity.

Social Relationships. This domain highlights personal relationships, social support, and sexual activity.

Environment. This domain consists of physical safety and security, home environment, financial resources, accessibility and quality of health and social care, opportunities for acquiring new information and skills, participation in and opportunities for recreation and leisure activities, the physical environment such as pollution, noise, traffic and climate, as well as transport facilities.

Spirituality/Religion/Personal Beliefs. This domain explores the role of spirituality, religion, and personal belief systems in shaping one's quality of life. It acknowledges the significance of individual values, faith, and philosophical perspectives that guide coping mechanisms and give meaning to life experiences.

WHOQOL-BREF - Description and Domains

The WHOQOL-BREF is a widely recognized instrument developed by the World Health Organization (WHO) as a brief yet comprehensive tool for assessing an individual's Quality of Life (QoL). It serves as a shortened version of the original WHOQOL-100, retaining the conceptual richness of the full questionnaire while offering greater practicality and efficiency, particularly in time-limited clinical and research settings.

Designed to be culturally adaptable, the WHOQOL-BREF is suitable for use across diverse populations, geographic regions, and social contexts. It is frequently used in both clinical practice (e.g., mental health assessments, chronic illness management) and academic research (e.g., population studies, public health evaluations).

Structure of the WHOQOL-BREF

The WHOQOL-BREF consists of 26 items in total, with 2 general questions, one on the overall perception of an individual's quality of life and the other on their overall perception

of health. There are 24 specific questions grouped into 4 key domains that reflect essential dimensions of well-being.

Physical Health. This domain assesses the physical well-being of an individual, covering aspects such as pain and discomfort, levels of energy and fatigue, and sleep and rest. These indicators reflect the extent to which physical health enables or restricts daily functioning and overall quality of life (World Health Organization [WHO], 1996).

Psychological Health. The psychological domain focuses on mental and emotional well-being. It includes positive feelings, cognitive processes such as thinking, learning, memory, and concentration, along with self-esteem and body image. Negative emotions like anxiety, sadness, or depression are also considered, making this domain crucial in understanding psychological resilience (Skevington, Lotfy, & O'Connell, 2004). This domain is essential for understanding how stress, emotional health, and personal outlook contribute to overall well-being.

Social Relationships. The social relationships domain highlights the importance of interpersonal and social interactions. It considers personal relationships, availability of social support, and aspects of sexual activity. These elements contribute significantly to feelings of belonging, connectedness, and emotional support (Harper & Power, 1998).

Environment. The environmental domain evaluates external factors that influence quality of life. It includes perceptions of physical safety and security, home environment, financial resources, and access to health and social care. Additionally, it assesses opportunities for learning, recreation, and leisure, as well as the influence of the physical environment (pollution, noise, climate) and transport facilities (WHO, 1996; Skevington et al., 2004).

Applications and Relevance

The WHOQOL-BREF is widely utilized to evaluate quality of life among working mothers, offering insights into how different occupational settings, such as work-from-home (WFH) versus onsite employment, affect multiple dimensions of well-being. For example,

Physical health may be influenced by commute time, ergonomic setups, or physical activity levels. Psychological health may shift due to isolation, multitasking stress, or work-life boundaries. Social relationships can be impacted by changes in daily social interactions or family dynamics. Environmental factors may vary with job flexibility, access to childcare, or financial stability.

In the post-pandemic work culture, where hybrid and remote work models have become more common, the WHOQOL-BREF is an invaluable tool for identifying shifts in well-being and guiding policies or interventions that support work-life balance and employee health.

The WHOQOL-BREF is widely used to assess QoL among working mothers, helping researchers understand how different work arrangements, like work-from-home (WFH) vs on-site jobs, influence their physical, emotional, social, and environmental well-being. Its structured format makes it easier to compare the impact of occupational settings on daily life, especially in the post-pandemic work culture.

Domain-Wise Comparative Focus

Working mothers show distinct differences in physical, psychological, social, and environmental well-being based on whether they work onsite or from home (Kim, Ferrie, & Stone, 2003; Picker-Roesch, Schweiker, Kraus, & Lang, 2024).

Physical Health. On-site working mothers benefit from routine and sleep consistency but face fatigue, long commutes, and sometimes lack flexibility for health needs (Kim et al., 2003). Working from home offers more flexibility, fewer commutes, and ergonomic comfort, but can result in physical strain due to multitasking and more sedentary behaviors (Picker-Roesch et al., 2024).

Psychological Well-being. On-site work allows mothers direct social interaction and focus on tasks, which may enhance self-worth, but they also face stress balancing work and family (Kim et al., 2003). Mothers working from home feel emotionally secure and enjoy control over their day, yet are at risk of cognitive overload and loneliness due to blurred boundaries and reduced peer contact (Picker-Roesch et al., 2024).

Social Relationships. On-site working mothers experience community and professional networking but may have less time with family, affecting personal relationships (Kim et al., 2003). At home, mothers have more family time and shared routines but may lack external social support and face interpersonal conflicts due to limited personal space (Kim et al., 2003).

Environment. On-site workspaces provide structure, safety, resources, and support, but long commutes and external stressors may challenge well-being (Picker-Roesch et al., 2024). Home-based mothers can personalize their environment, enjoying privacy and fewer stressors, though productivity may be affected by internet reliability, distractions, and socio-economic differences (Picker-Roesch et al., 2024).

Both work modes offer unique advantages and challenges, and neither is universally superior; hybrid approaches and inclusive policies may best support working mothers' quality of life (Kim et al., 2003; Picker-Roesch et al., 2024).

A study analyzing variation in the regional development of India based on the Physical Quality of Life Index, conducted by Somnath Mukherjee (2004), proposes that the index is based on three indicators, namely literacy rate, life expectancy, and infant mortality rate. Variations among the states show the uneven distributions of these indicators. States like Kerala, Goa, and Tripura have almost reached the optimum level of development. Some states like Bihar, Jharkhand, Uttar Pradesh, and Madhya Pradesh with huge area extensions and populations showing a sluggish rate of development.

A study conducted by Pavan G, Kumar M, Murti K, Dhingra S, Ravichandiran V., aimed to assess the factors influencing health-related quality of life (HRQoL) in patients experiencing adverse drug reactions (ADRs) at a tertiary care public sector hospital. A cross-sectional study was conducted over a period of 18 months and included both male and female patients aged 18 years and above. Patients who visited the outpatient and inpatient departments with complaints associated with ADRs were included in this study. HRQoL data were collected using the EuroQol-5 Dimension-5 Level (EQ-5D-5L) questionnaire to assess five dimensions of health on a five-level scale. Descriptive statistics, t-tests, and analysis of variance were used to analyse the data. The majority (68%) of the patients had moderately severe ADRs, and 63% of the participants had an income < 2.5 lakh Indian rupees (3009 USD). The mean EQ-5D-5L and EuroQoL. The variables ADR severity, income, and age showed a significant difference ($p < 0.05$) in HRQoL.

A systematic review by Rajesh Gondalia, Jagruti D. Parmar explores factors shaping the quality of life in pregnant women. Sociodemographic, physical, and psychological dimensions intertwine, demanding a holistic approach to maternal care. The present study contributes to improved maternal outcomes, ensuring that the journey through pregnancy is characterized by enhanced quality of life and holistic support.

Literature suggests that adolescents in rural India have reported better perceived QoL (Singh and Junnarkar 2014) and happiness (Sharma and Gulati 2012) as compared to their urban counterparts. In a study on predominantly urban school children from North India, Singh and Junnarkar (2014) used WHOQOL-BREF and found that all four domains of QoL significantly predicted positive mental health and well-being of the participants. Based on studies involving rural women, while some have reported lower QoL in rural areas as compared to urban areas (e.g., Mudey et al. 2011; Verma 2008), other studies like Bhatt (2013) found no significant difference in well-being between rural and urban women.

Further, education is strongly associated with health-related quality of life (HRQoL; D'Souza et al. 2013) as well as greater subjective and psychological well-being (Singh et al. 2014). Using WHOQOL-BREF with elderly people in urban Puducherry, Kumar et al. (2014) reported that QoL among the elderly was average. However, QoL was found to be significantly low among participants who were uneducated, living alone, and suffering from various health concerns. In a similar study involving rural elderly populations from North India, Qadri et al. (2013) reported that the majority of the participants (68.2%) enjoyed good QoL, with males experiencing better QoL in physical, social, psychological, and environmental domains. In a study comparing QoL among rural and urban elderly populations in North India, Kumari et al. (2018) suggested that urban participants reported significantly better QoL as compared to their peers from rural areas.

Pradhan (2008) carried out an interesting investigation wherein he studied the QoL index across India by applying totally fuzzing analysis (TFA), taking into account dimensions such as income, education, health, employment, and infrastructure. As per his empirical findings, QoL was substantially high in Indian states including Goa, Kerala, Punjab, Tamil Nadu, Maharashtra, Haryana, Himachal Pradesh, and Mizoram. However, QoL was found to

be quite low in Bihar, Odisha, Madhya Pradesh, Uttar Pradesh, and Assam. The remaining Indian states exhibited average levels of QoL.

The global rise of work-from-home (WFH) arrangements, particularly during and after the COVID-19 pandemic, has significantly altered the professional and personal lives of working mothers. While remote work offers flexibility and convenience, it also blurs boundaries between work and home, impacting quality of life (QoL). Peer-reviewed journals and global reports from 2018–2024 were sourced from databases including PubMed, JSTOR, ScienceDirect, and Google Scholar using keywords such as “work-from-home mothers,” “onsite working women,” “quality of life,” “remote work,” and “motherhood and employment.” Studies from multiple regions, including Europe, Asia, Africa, and North America, were selected to ensure a broad international perspective. This study uses a narrative literature review approach synthesizing international research comparing the QoL of work-from-home (WFH) and onsite working mothers across psychological, physical, social, and environmental domains.

Psychological Well-being. Remote work has been associated with both improvements and declines in psychological well-being, depending on context. For instance, Chung et al. (2021) in the UK found that flexible remote work arrangements contributed to higher life satisfaction and lower psychological distress among mothers with caregiving responsibilities. However, Wang et al. (2021), in a study across 10 countries including the US, China, and Germany, observed that high levels of work-home interference led to emotional exhaustion and reduced job satisfaction for WFH mothers.

In contrast, on-site working mothers often face rigid schedules that limit personal autonomy. López-Igual and Rodríguez-Modroño (2020), using EU-LFS data across 28

European countries, reported that onsite working mothers exhibited higher stress levels and emotional strain, especially when childcare support was inadequate.

Physical Health. A study in Australia by Oakman et al. (2020) highlighted that WFH mothers experienced reduced commuting-related fatigue and more opportunities for sleep and exercise. Similarly, Koh, Allen, and Zafar (2021) in Singapore found that WFH improved physical health markers due to better time management and fewer exposure risks during the pandemic. However, long-term remote work may lead to sedentary habits and poor ergonomics. Zhang et al. (2021) in China noted that remote-working women experienced increased musculoskeletal issues due to non-ideal home setups. In comparison, on-site working mothers often maintain a more structured routine, but face higher physical exhaustion. Marmot et al. (2018), analyzing occupational health data in Canada and the UK, reported a higher incidence of stress-related ailments among mothers juggling long working hours and household responsibilities.

Social Relationships. Social connectedness is a vital component of QoL. Nolting et al. (2022) in Germany found that WFH led to feelings of isolation among women, particularly those with limited adult interaction outside the household. Similarly, Agarwal and Lenka (2020) in India observed that the absence of social support systems intensified feelings of loneliness for remote-working mothers. By contrast, on-site working mothers benefit from in-person peer interaction, enhancing their sense of social identity. A study by Yildirim and Eslen-Ziya (2020) in Turkey highlighted that office-based employment offered mothers emotional relief through workplace socialization, although it came at the cost of reduced time with children.

Environmental Factors. Environmental satisfaction varies greatly between WFH and onsite setups. Chung and van der Lippe (2020) in the Netherlands found that remote-working

mothers enjoyed greater autonomy and home comfort but struggled with household distractions, especially in multi-generational homes. In contrast, Nguyen and Armoogum (2021) in France and Vietnam reported that poor digital infrastructure and space constraints were major stressors in WFH environments. On-site environments, although more controlled and structured, often fail to accommodate the needs of mothers, particularly in patriarchal or rigid corporate cultures. Kanwar et al. (2021) in the UAE documented how the lack of lactation rooms, childcare support, and rigid work hours in offices negatively affected mothers' environmental QoL.

Cultural and Policy Influences. Cross-cultural differences significantly shape mothers' QoL. Biroli et al. (2021) compared Italy, Spain, and the UK, finding that national family policies, gender norms, and availability of flexible working arrangements were major determinants of maternal well-being. Scandinavian countries like Norway and Sweden, with progressive leave policies and subsidised childcare, consistently report higher QoL scores among both WFH and onsite working mothers (Hjálmsdóttir & Bjarnadóttir, 2021). In low- and middle-income countries, lack of structural support often exacerbates the challenges. Akinwale (2020) in Nigeria found that without employer flexibility or domestic help, onsite working mothers had the lowest QoL scores across all domains.

International literature reveals that both WFH and onsite work have unique benefits and stressors that affect mothers' quality of life. Remote work offers autonomy and convenience but risks emotional isolation and physical strain without proper boundaries. On-site work ensures structure and social contact but increases time pressure and stress, especially without supportive policies. A hybrid work model, underpinned by gender-sensitive workplace reforms and cultural flexibility, may offer the most promising path forward. The review suggests no absolute superiority between WFH and onsite models.

Instead, quality of life outcomes depend on factors like autonomy, support systems, infrastructure, and cultural norms. While WFH provides time flexibility and comfort, it risks mental fatigue and role conflict. On-site work offers structure and social engagement, but often at the cost of stress and time imbalance. Countries with supportive parental leave, childcare access, and workplace flexibility enable better QoL for both groups. Intersectional factors like marital status, socio-economic class, and number of children further influence these outcomes.

The shift in work modalities has created a complex landscape for working mothers. Both work-from-home and onsite models bring advantages and stressors. International evidence suggests that institutional support, gender-sensitive workplace policies, and flexible hybrid systems are essential for improving maternal QoL in both setups.

Aim: To compare the quality of life (QoL) between work-from-home (WFH) and onsite working mothers using the WHOQOL-BREF (World Health Organization Quality of Life - BREF) questionnaire. The study aims to assess the impact of different work environments on the physical, psychological, social, and environmental well-being of working mothers.

METHODOLOGY

Preliminaries

Study Design: The study follows a cross-sectional comparative design to evaluate and compare the quality of life between two groups of working mothers: those who work from home (WFH) and those who work onsite (in the office).

Participants: A sample of 36 working mothers (18 with work-from-home jobs and 18 on-site jobs) with an age range of 35-55 years was chosen from the urban population of Kolkata, West Bengal.

Inclusion Criteria: Mothers, employed for a minimum of 6 months either in WFH or on-site work conditions, between the ages of 35 and 55 years.

Exclusion Criteria: Mothers who are currently unemployed or on maternity leave.

Materials Required

1. **WHOQOL-BREF Questionnaire:** The WHOQOL-BREF is a standardized tool that consists of 26 items. It is designed to measure the following domains of Quality of Life:
 - Physical Health: Includes aspects like pain, sleep quality, and mobility.
 - Psychological Health: Includes aspects like positive emotions, self-esteem, and concentration.
 - Social Relationships: Includes personal relationships, social support, and sexual activity.
 - Environmental Health: Includes physical safety, financial resources, and access to healthcare.
2. **Informed Consent Form:** A consent form will be required from all participants to ensure ethical standards are met and privacy is maintained.
3. **Demographic Questionnaire:** A short survey collecting basic demographic details like age, marital status, number of children, and work sector.
4. **Data Collection Tools:** Survey Software (for online responses, if applicable) or paper forms for offline responses.

Description of the Test Manual

The WHOQOL-100 allows a detailed assessment of each facet relating to quality of life. In certain instances, however, the WHOQOL-100 may be too lengthy for practical use. The

WHOQOL BREF Field Trial Version has therefore been developed to provide a short-form quality of life assessment that looks at domain-level profiles, using data from the pilot WHOQOL assessment and all available data from the Field Trial Version of the WHOQOL-100. Twenty field centres situated within eighteen countries have included data for these purposes. The WHOQOL-BREF contains a total of 26 questions. To provide a broad assessment, one item from each of the 24 facets contained in the WHOQOL-100 has been included. In addition, two items from the Overall Quality of Life and General Health facet have been included.

The WHOQOL-BREF is available in 19 different languages. The appropriate language version and permission for using it can be obtained from The WHOQOL Group, Programme on Mental Health, World Health Organisation, CH-1211 Geneva 27, Switzerland. Under no circumstances should the WHOQOL-BREF be used without consultation with the WHOQOL Group. A methodology has been developed for new centres wishing to develop a further language version of the WHOQOL-100 or the WHOQOL-BREF. This can be obtained from The WHOQOL Group, Programme on Mental Health, World Health Organisation, CH-1211, Geneva 27, Switzerland. Questions should appear in the order in which they appear in the example WHOQOL-BREF provided within this document, with instructions and headers unchanged. Questions are grouped by response format. The equivalent numbering of questions between the WHOQOL-BREF and the WHOQOL-100 is given in the example version of the WHOQOL-BREF to enable easy comparison between responses to items on the two versions. The WHOQOL-100 field test permitted centres to include national items or facets that were thought to be important in assessing quality of life. Where centres wish to include additional national items or modules to the WHOQOL-BREF, these should be included on a separate sheet of paper and not scattered amongst the existing 26 items. There are three reasons for this: to control for item order effects, which could occur and change

item meaning, the WHOQOL-BREF represents an agreed-upon core set of international items, and the WHOQOL-BREF is likely to be used where quality of life is one of several parameters being assessed. Therefore, additional national information can be obtained by including additional modules and measures.

Administration of the WHOQOL-BREF

New centres using the WHOQOL-BREF must follow the same field testing procedure as used for the WHOQOL-100, including piloting the instrument on at least 300 culturally-defined adult participants. The sample should follow a quota: 50% under 45 years and 50% aged 45 or older; 50% male and 50% female; and 250 individuals with a disease or impairment, alongside 50 healthy individuals. The diseased group should represent varied levels of quality of life, including those with chronic or transient conditions, sampled from primary care, hospitals, and community settings. The WHOQOL-BREF should be self-administered when possible, or interviewer-assisted if needed, using standardized instructions. While previous WHOQOL-100 centers are encouraged to follow this process, it's not mandatory for specific patient studies. The default reference period is two weeks, though this may be extended (e.g., to four weeks) for chronic conditions or adapted for cultural contexts.

Procedure

The administration of the WHOQOL-BREF was conducted either online for mothers working from home or through a paper-based form for those working onsite, with similar conditions ensured to minimize environmental bias. The questionnaire was self-administered, and participants received clear instructions on how to respond to each item. For every question, participants rated their satisfaction with different aspects of life using a 5-point Likert scale,

ranging from 1 (very dissatisfied) to 5 (very satisfied). Completion of the WHOQOL-BREF took approximately 10–15 minutes. Informed consent was from all the participants and they were assured that their responses would remain anonymous and confidential, and that the results would be used exclusively for research purposes.

Instructions

This assessment asks how you feel about your quality of life, health, or other areas of your life. Please answer all the questions. If you are unsure about which response to give to a question, please choose the one that appears most appropriate. This can often be your first response. Please keep in mind your standards, hopes, pleasures and concerns. We ask that you think about your life in the last two weeks. For example, thinking about the last two weeks, a question might ask: Do you get the kind of support from others that you need? So the options are [1. Not at all, 2. Not much, 3. Moderately, 4. A great deal, 5. Completely} 1 2 3 4 5 You should circle the number that best fits how much support you got from others over the last two weeks. So you would circle the number 4 if you got a great deal of support from others, as follows. You would circle number 1 if you did not get any of the support that you needed from others in the last two weeks. Please read each question, assess your feelings, and circle the number on the scale for each question that gives the best answer for you.

Scoring Procedure

The WHOQOL-BREF (Field Trial Version) produces a quality of life profile. It is possible to derive four domain scores. There are also two items that are examined separately: question 1 asks about an individual's overall perception of quality of life, and question 2 asks about an individual's overall perception of their health. The four domain scores denote an individual's perception of quality of life in each particular domain. Domain scores are scaled in a positive

direction (i.e. higher scores denote a higher quality of life). The mean score of items within each domain is used to calculate the domain score. Mean scores are then multiplied by 4 in order to make domain scores comparable with the scores used in the WHOQOL-100.

A method for the manual calculation of individual scores is given on page 1 of the WHOQOL-BREF assessment form. The method for converting raw scores to transformed scores when using this method is given. The first transformation method converts scores to a range between 4-20, comparable with the WHOQOL-100. The second transformation method converts domain scores to a 0-100 scale

Any national items should be scored separately from the core 26 items of the BREF. During the analysis, the performance of any national items will be examined for possible use in alter national studies. At this stage of field testing, national and core items must not be mixed inthe administration or scoring of the BREF.

1. Norms

WHO QOL, Bref- Preliminary Norms: Hawthorne, Herman& Murphy (2006)

WHO QOL- Bref Domains	Physical Health	Psychological Health	Social Relationships	Environment
Mean	73.5	70.6	71.5	75.1
SD	18.1	14.0	18.5	13.0

2. Reliability & Validity

The WHOQOL-BREF has consistently demonstrated strong psychometric properties across a variety of cultural and clinical settings. Studies have shown that the instrument has good internal consistency, with Cronbach's alpha values typically exceeding .70 in all domains, physical, psychological, social, and environmental (Skevington et al., 2004; Oliveira et al., 2016). Test-retest reliability has also been found to be satisfactory, with intraclass correlation coefficients (ICCs) ranging from .76 to .80 (Su et al., 2006). Construct validity is supported by confirmatory factor analyses that validate the four-domain structure across diverse populations (Skevington et al., 2004). The tool also exhibits strong convergent and criterion validity, demonstrated through significant correlations with other measures of quality of life such as the SF-36, and its ability to differentiate between individuals with and without chronic conditions (Wong et al., 2019). Additionally, Rasch analysis has confirmed the instrument's unidimensionality and measurement precision, although the social domain occasionally shows lower reliability due to fewer items (Thitiwan et al., 2023). Overall, the WHOQOL-BREF is a reliable and valid tool for assessing quality of life across different settings and populations.

Results

The study compared the quality of life between women working from home and those working onsite, examining four WHOQOL-BREF domains: physical health, psychological health, social relationships, and environment. Using independent-samples t-tests and Mann-Whitney U tests to analyze data from 19 participants in each group, no significant differences were found across any domains despite small variations in mean scores. These findings suggest that work location alone does not significantly impact quality of life, highlighting the role of other factors like support systems and personal coping in shaping wellbeing.

Table 1

Raw Scores of remote and onsite working mothers based on overall perception of quality of life (QOL), overall perception of general health, physical, psychological, social & environmental.

Working Mothers Categories	Overall Perception of General					
	Overall perception of QOL	Health	Physical	Psychological	Social	Environmental
	4	2	24	18	10	25
	3	4	24	20	10	26
	2	2	18	17	14	25
	3	3	26	21	10	24
	2	2	19	14	7	13
REMOTE	3	4	19	19	12	21
	3	4	21	19	12	26
	3	2	19	12	13	31
	4	4	21	20	9	23
	3	2	19	12	13	31
	3	4	22	19	6	15

	3	2	19	12	13	31
	3	4	22	19	6	15
	4	5	26	20	10	34
	4	4	25	26	12	34
	4	3	22	21	10	25
	4	4	24	22	9	33
	3	4	22	20	9	18
	3	2	19	12	13	31
	4	3	21	18	10	19
	4	2	24	16	10	22
	4	4	29	23	15	15
	4	3	19	19	13	25
	4	4	28	25	14	32
ONSITE	4	3	28	25	11	28
	3	4	26	21	10	27
	2	4	24	18	10	24
	2	3	20	15	9	27
	4	4	20	20	9	26

2	2	17	9	6	18
4	4	20	20	9	26
2	2	17	9	6	18
4	4	27	22	9	31
4	3	24	21	11	32
4	5	23	22	11	27
4	4	19	23	14	30
4	4	22	21	12	29
2	2	19	9	8	21

The table displays the raw scores collected from individual remote and onsite working mothers for their overall perception of quality of life, general health, and four key domains: physical health, psychological wellbeing, social relationships, and environmental factors. These are the direct responses gathered from the WHOQOL-BREF questionnaire, reflecting each participant's self-assessment before any transformations or standardizations were applied.

Table 2

Transformed scores of remote and onsite working mothers based on physical, psychological, social, & environmental in scale on 4-20 and 0-100 as per WHO-QOL.

Working Mothers	Physical		Psychological		Social		Environmental	
	4-20	0-100	4-20	0-100	4-20	0-100	4-20	0-100
	14	63	12	50	13	56	13	56
	14	63	13	56	13	56	13	56
	10	38	11	44	19	94	13	56
	15	69	14	63	13	56	12	50
	11	44	9	31	9	31	7	19
	11	44	13	56	16	75	11	44
Remote	12	50	13	56	16	75	13	56
	11	44	8	25	17	81	16	75
	12	50	13	56	12	50	12	50
	11	44	8	25	17	81	16	75
	13	56	13	56	8	25	8	25
	11	44	8	25	17	81	16	75
	13	56	13	56	8	25	8	25

	15	69	13	56	13	56	17	81
	16	75	17	81	16	75	17	81
	13	56	14	63	13	56	13	56
	14	63	15	69	12	50	17	81
	13	56	13	56	12	50	9	31
	11	44	8	25	17	81	16	75
Onsite	12	50	12	50	13	56	10	38
	14	63	11	44	13	56	11	44
	17	81	15	69	69	29	69	25
	11	44	13	56	17	81	13	56
	16	75	17	81	19	94	16	75
	16	75	17	81	15	69	14	63
	15	69	14	63	13	56	14	63
	14	63	12	50	13	56	12	50
	11	44	10	38	12	50	14	63
	11	44	13	56	12	50	13	56
	10	38	6	13	8	25	9	31
	11	44	13	56	12	50	13	56

10	38	6	13	8	25	9	31
15	69	15	69	12	50	16	75
14	63	14	63	15	69	16	75
13	56	15	69	15	69	14	63
11	44	15	69	19	94	15	69
13	56	14	63	16	75	15	69
11	44	6	13	11	44	11	44

The table demonstrates the transformed scores for remote and onsite working mothers in physical, psychological, social, and environmental domains. These scores are shown on both a standardized 4–20 scale and a 0–100 scale, allowing easier comparison across participants and between domains. Higher scores on these scales indicate better perceived quality of life in each specific domain, in line with WHOQOL-BREF conventions.

Table 3

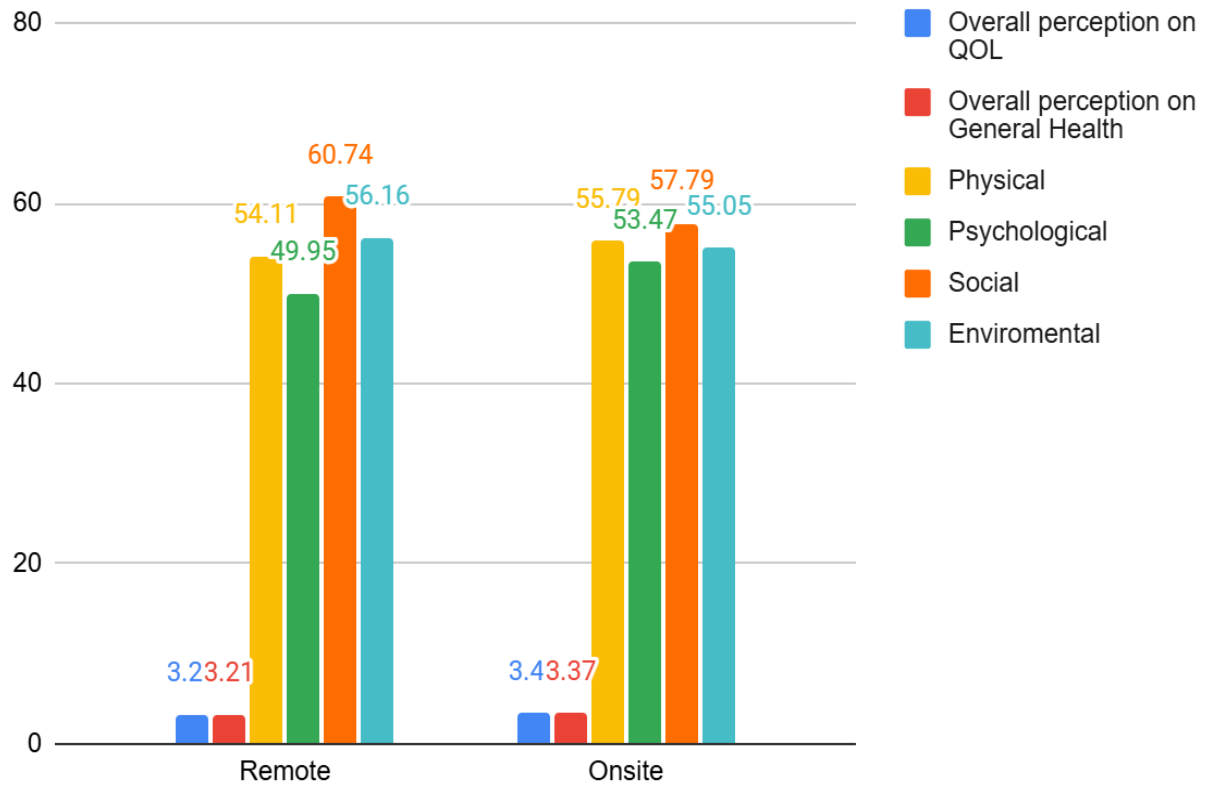
Mean and SD of Remote and Onsite working mothers based on overall perception on quality of life (QOL), overall perception on general health, physical, psychological, social & environmental.

Working Mothers Categories	Overall perception of QOL		Overall Perception of General Health		Physical		Psychological		Social		Environmental	
	Mean	SD	n	SD	n	SD	n	SD	n	SD	Mean	SD
Remote	3.21	0.63	3.21	1.03	54.11	10.64	49.95	16.41	60.74	20.08	56.16	20.34
Onsite	3.42	0.87	3.37	1.00	55.79	13.73	53.47	19.09	57.79	20.32	55.05	18.35

The table presents the mean values and standard deviations for various quality of life indicators among remote and onsite working mothers. It includes the overall perception of quality of life (QOL), general health, and the four domains measured by the WHOQOL-BREF: physical, psychological, social, and environmental. Remote working mothers reported a mean overall QOL of 3.21 with a standard deviation of 0.63, and a similar mean of 3.21 (SD = 1.03) for general health. Their scores across the domains were 54.11 (SD = 10.64) for physical, 49.95 (SD = 16.41) for psychological, 60.74 (SD = 20.08) for social, and 56.16 (SD = 20.34) for environmental. In comparison, onsite working mothers showed a slightly higher mean overall QOL of 3.42 (SD = 0.87) and general health mean of 3.37 (SD = 1.00). Their domain scores were similarly close: physical domain at 55.79 (SD = 13.73), psychological at 53.47 (SD = 19.09), social at 57.79 (SD = 20.32), and environmental at 55.05 (SD = 18.35). This table provides a concise summary of the average experiences and the variability within each group, showing generally similar quality of life perceptions regardless of work setting.

Figure 1

Graphical Representation of comparison of the mean of remote and onsite working mothers based on physical, psychological, social & environmental domains.

**Table 4**

Independent sample Mann-Whitney U test of remote and onsite working mothers based on **physical, psychological, social, & environmental** domains.

Null Hypothesis	Test	Sig.	Decision
The distribution of Physical is the same across categories of working women(WM)	Independent Sample Mann-Whitney U Test	.817	Retain the Null Hypothesis

The distribution of Psychological is the same across categories of working women(WM)	Independent Sample Mann-Whitney U Test	.370	Retain the Null Hypothesis
The distribution of Social is the same across categories of working women(WM)	Independent Sample Mann-Whitney U Test	.506	Retain the Null Hypothesis
The distribution of Environmental is the same across categories of working women(WM)	Independent Sample Mann-Whitney U Test	.773	Retain the Null Hypothesis

The table details the results of the Mann-Whitney U test conducted to compare the scores from remote and onsite working mothers in each domain statistically. For the physical domain, the significance value (p) is 0.817, which is well above the cutoff of 0.05, leading to the retention of the null hypothesis that there is no difference between groups. Similarly, the psychological domain yielded a p-value of 0.370, the social domain 0.506, and the environmental domain 0.773, all of which exceed the threshold for statistical significance. As a result, for all four domains, the null hypothesis, stating the distributions are the same across remote and onsite working mothers, was retained. This statistical result confirms that quality of life scores do not differ significantly between mothers working remotely and those working onsite across any domain, indicating comparable experiences in physical, psychological, social, and environmental aspects of quality of life between the two groups.

Interpretation

The present study aims to compare the quality of life of working mothers in context of remote and onsite working conditions by administering the WHOQOL-BREF (abbreviated version of WHOQOL).

Quality of life is defined as “an individual's perception of their position in life in the context of the culture and value systems in which they live and in relation to their goals, expectations, standards, and concerns” (World Health Organization, 1997). Quality of life (QOL) is an essential indicator in health and medical research, as it provides valuable insight into how illness, treatment, and broader social circumstances affect an individual's well-being and daily functioning.

Item 1 measures an individual's overall perception of their quality of life. According to the results obtained, the mean and SD for remote working mothers is found to be only slightly higher than the mean and SD for onsite working mothers and it can be interpreted that the overall perception of quality of life is moderate for both the groups. Item 2 measures an individual's overall perception of their health. According to the results obtained, the mean and SD for remote working mothers is found to be slightly higher than the mean and SD for onsite working mothers and thus the overall perception of health is also found to be moderate for both the groups. Such findings are similar with the studies of Wang et al. (2021), which observed that high levels of work-home interference led to emotional exhaustion and reduced job satisfaction for WFH mothers. While remote work offers flexibility and convenience, it also blurs boundaries between work and home, impacting quality of life (QoL).

In the physical domain, the results obtained suggest that the perceived physical health measured on the parameters of energy level, mobility, dependence on medical aids, sleep and work capacity is found to be higher for onsite working mothers in

comparison to remote working mother. But however when statistically testing it , no significant difference was found between the perceived physical health of remote and onsite working mothers. This lack of significant difference can be supported by prior similar studies; such as Zhang et al. (2021) found that remote-working women experienced increased musculoskeletal issues due to non-ideal home setups ,with caregiving responsibilities amplifying discomfort.

In the psychological domain , the mean and SD of the transformed scores obtained by remote working mothers is found to be slightly higher than the mean and SD for onsite working mothers. Thus the psychological health is found to be slightly higher for onsite working mothers in comparison to remote working mothers and the facets on which they have scored high are self esteem, bodily image, presence of positive feelings, memory and concentration , personal religious and spiritual beliefs. But when statistically tested , no significant difference is found between the psychological health of remote and onsite working mothers. These findings are consistent with Hall et al. (2023), whose systematic review reported that enforced home working was often linked with poorer mental health (stress, anxiety, psychological distress), especially for women with young children or crowded homes; feelings of isolation repeatedly mediated worse outcomes versus on-site work with in-person contact.

In the social domain , the perceived social relationships of remote working mothers is found to be higher than onsite working mothers when measured on the facets of better social support , personal relationships and satisfaction with intimacy and sexual activity. However when statistically tested, no significant difference was found between the social relationships of remote and onsite working mothers. Consistent with this finding , similar research , for eg. Tu, Weng & Hsu (2022) has also stated that WFH improved family relationships by increasing time with family and enhancing work–life balance,

identifying concrete “adaptive processes” like boundary setting, routine coordination that helped parents—especially mothers—nurture family cohesion relative to prepandemic on-site norms.

In the environmental domain , remote working mothers have scored higher than onsite working mothers when measured on the facets of financial stability, sense of safety and personal security, accessibility and quality of healthcare and social services, better home environment , increased access to information and skill development , overall improved physical environment in terms of pollution , traffic, noise etc. However when statistically tested, no significant difference was found between the environmental factors of remote and onsite working mothers. Environmental satisfaction varies greatly between WFH and onsite setups and many studies have explored this area , such as , Chung and Lippe (2020) in the Netherlands found that remote-working mothers enjoyed greater autonomy and home comfort but struggled with household distractions, especially in multi-generational homes. Work from home arrangements reduces commuting costs, wardrobe expenses, and daily travel time ,there by improving financial stability and freeing resources for childcare, healthcare, and education (McLean Hospital, 2022)

Overall, the findings of the present study suggest that while onsite working mothers report slightly higher scores in overall quality of life, perceived physical and psychological health ; remote working mothers tend to show marginally better outcomes in social relationships and environmental satisfaction. However, none of these differences were found to be statistically significant, indicating that both working conditions offers a similar overall quality of life and any differences could be due to chance factors or individual differences rather than the type of work arrangement alone. Instead, quality of life is shaped more by **contextual factors** such as personal values,

autonomy, social support, work–life balance strategies, and access to resources which can vary widely among individuals in both groups.

CONCLUSION :

The present study found that both remote and onsite working mothers have similar moderate level of overall perceived quality of life. While onsite working mothers report slightly higher scores in perceived physical and psychological health ; remote working mothers scored slightly higher in social relationships and environmental satisfaction.

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